

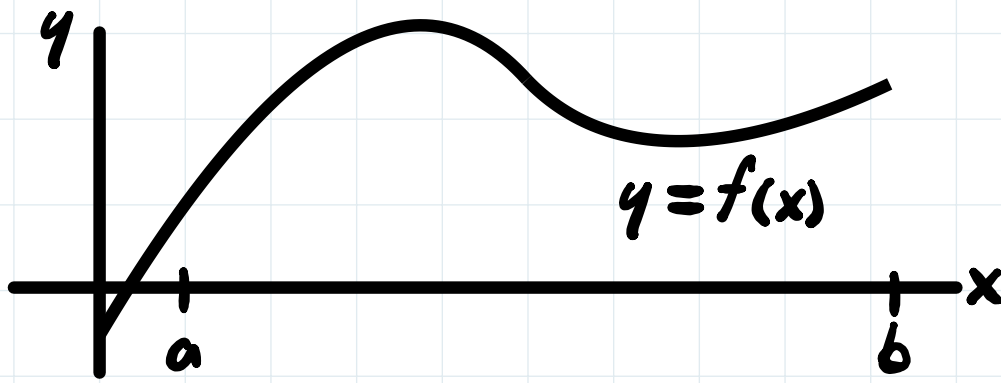
§8.1 - Arc Length

Problem

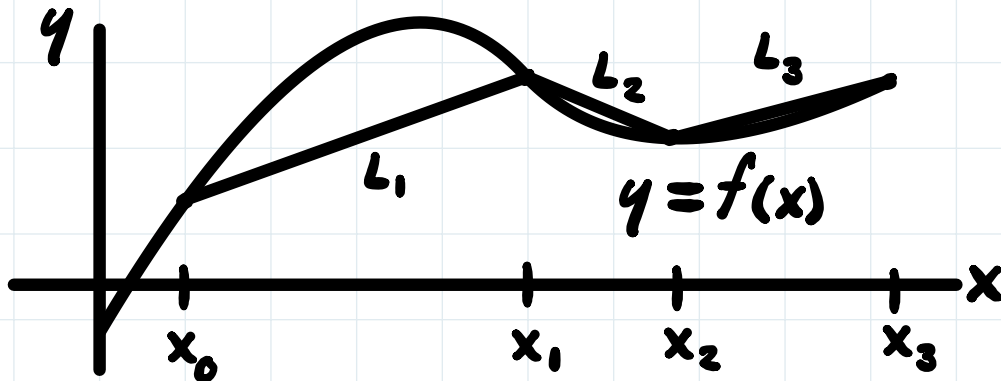
We want to measure the length of the curve given by

$$\{(x, f(x)) \mid x \in [a, b]\}$$

where f is a “smooth” function on $[a, b]$.



We assume the curve is piecewise linear.



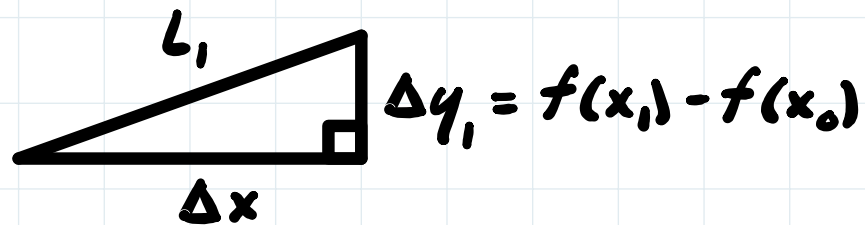
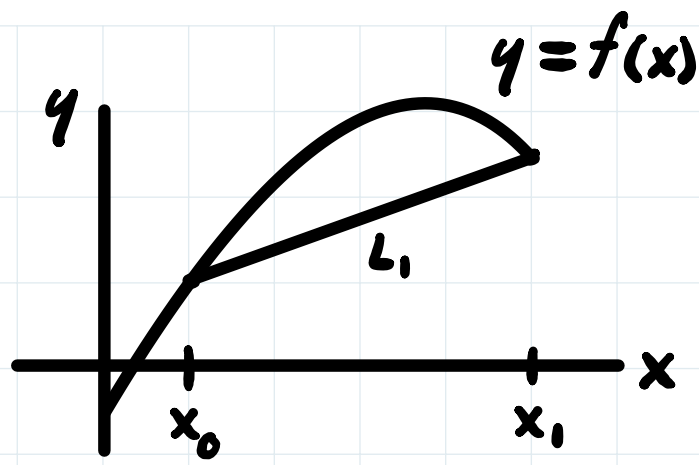
Partition $[a, b]$ with stepsize Δx

$$x_k = a + k\Delta x, \quad I_k = [x_{k-1}, x_k]$$

for $k = 0, 1, \dots, n$.

Estimate the length of the curve over I_k by L_k .

Consider $k = 1$:



Width of the triangle: Δx

Height of the triangle:

$$\Delta y_1 = f(x_1) - f(x_0)$$

then length L_1 satisfies

$$L_1^2 = (\Delta x)^2 + (\Delta y_1)^2$$

$$= (\Delta x)^2 \left(1 + \left(\frac{\Delta y_1}{\Delta x} \right)^2 \right)$$

$$= (\Delta x)^2 \left(1 + \left(\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right)^2 \right).$$

By the M.V.T.

$$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(\bar{x}_1)$$

$$\Rightarrow L_1 = (1 + (f'(\bar{x}_1))^2)^{1/2} \Delta x$$

where $\bar{x}_1 \in [x_0, x_1]$.

Add up all of the approximations

$$L \approx \sum_{k=1}^n L_k = \sum_{k=1}^n (1 + (f'(\bar{x}_k))^2)^{1/2} \Delta x$$

and let $n \rightarrow \infty$

$$L = \int_a^b (1 + (f'(x))^2)^{1/2} dx.$$

Computing L is often hard because so few functions make it easy to integrate

$$(1 + (f'(x))^2)^{1/2}.$$

Mean Value Theorem

If f is a differentiable function on the interval $[a, b]$, then there exists a number c between a and b such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

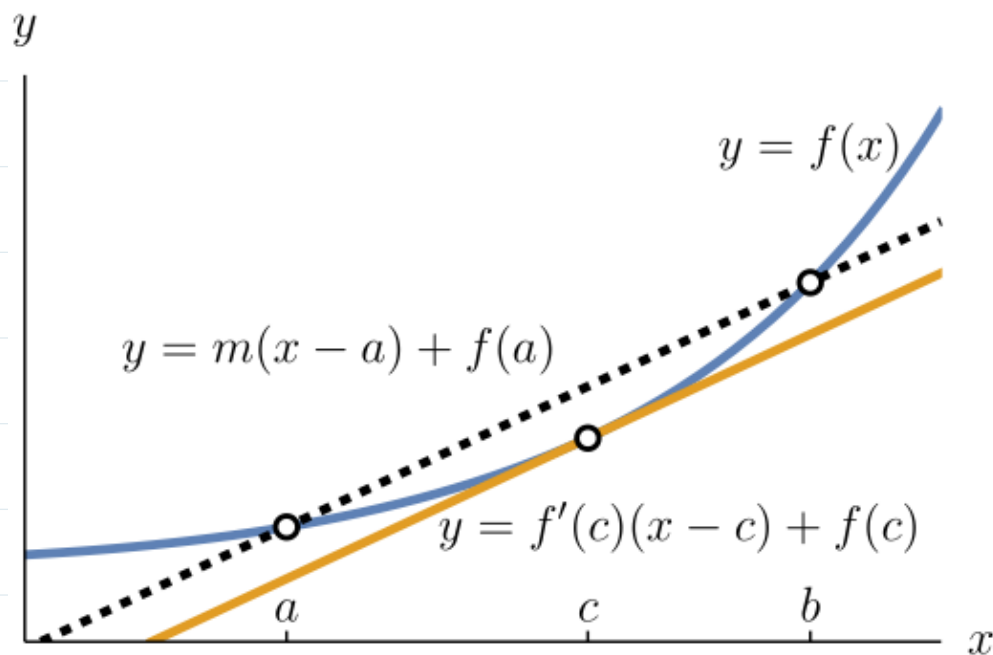
or, equivalently,

$$f(b) - f(a) = f'(c)(b - a).$$

This means we have a secant line, with a slope

$$\frac{f(b) - f(a)}{b - a}$$

***that is parallel with some tangent line of f .
The slope of the secant line is f 's average
rate of change over $[a, b]$.***



Theorem #1

Let f' be continuous on $[a, b]$.

Then the length of the curve from $(a, f(a))$ to $(b, f(b))$ is given by

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx.$$

We call L the **arc length** of f over the interval $[a, b]$.

Example #1

Find the arc length of

$$y = \frac{2}{3}x^{3/2}$$

over $[0, 3]$.

Here

$$f'(x) = x^{1/2}$$

so

$$(1 + (f'(x))^2)^{1/2} = (1 + x)^{1/2}.$$

Then

$$\begin{aligned} L &= \int_0^3 (1+x)^{1/2} dx \\ &= \frac{2}{3} (1+x)^{3/2} \Big|_{x=0}^3 \\ &= \frac{2}{3} (4^{3/2} - 1) \\ &= \frac{2}{3} (2^3 - 1) \\ &= \frac{14}{3} \end{aligned}$$

Example #2

Find the arc length of

$$y = \frac{e^x}{2} + \frac{e^{-x}}{2}$$

over $[-\ln(2), \ln(2)]$.

Here

$$f'(x) = \frac{e^x}{2} - \frac{e^{-x}}{2}$$

while

$$(f'(x))^2 = \left(\frac{e^x}{2} - \frac{e^{-x}}{2}\right)\left(\frac{e^x}{2} - \frac{e^{-x}}{2}\right)$$

$$= \frac{e^{2x}}{4} - 2 \frac{e^x e^{-x}}{4} + \frac{e^{-2x}}{4}$$

$$= \left(\frac{e^x}{2}\right)^2 - \frac{1}{2} + \left(\frac{e^{-x}}{2}\right)^2$$

$$\Rightarrow 1 + (f'(x))^2 = \left(\frac{e^x}{2}\right)^2 + \frac{1}{2} + \left(\frac{e^{-x}}{2}\right)^2$$

$$= \left(\frac{e^x}{2} + \frac{e^{-x}}{2}\right)^2$$

how convenient...

$$(1 + (f'(x))^2)^{1/2} = \frac{e^x}{2} + \frac{e^{-x}}{2}$$

$$\int_{-\ln(2)}^{\ln(2)} \frac{e^x}{2} + \frac{e^{-x}}{2} dx = \frac{e^x}{2} - \frac{e^{-x}}{2} \Big|_{x=-\ln(2)}^{\ln(2)}$$

$$= \frac{1}{2} \left(2 - \frac{1}{2}\right) - \frac{1}{2} \left(\frac{1}{2} - 2\right)$$

$$= \frac{3}{2}.$$

Definition #1

Let f' be continuous on $[a, b]$.

Then

$$s(x) = \int_a^x \sqrt{1 + (f'(t))^2} \, dt$$

for $x \in [a, b]$ is the **arc length function** of f .

We have

$$s'(x) = \sqrt{1 + (f'(x))^2}$$

and since

$$1 + (f'(x))^2 \geq 1$$

we have

$$s'(x) \geq 1.$$

The only time $s'(x) = 1$ is whenever $f'(x) = 0$:

1. f' is constant along an interval
2. f' , the slope of the curve, is 0

Example #3

A hawk flying at 15 m/s at an altitude of 180 m accidentally drops its prey. The path of the falling prey is described by the equation

$$y = 180 - \frac{x^2}{45}$$

until it hits the ground, where y is its height above the ground and x is the horizontal distance traveled in meters.

Find the integral that measures the distance traveled by the prey from the time it is dropped until the time it hits the ground

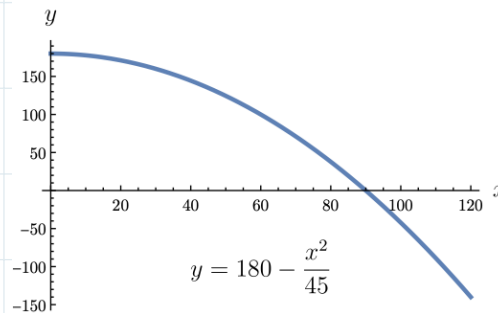
The prey hits the ground when $y = 0$,

$$0 = 180 - \frac{x^2}{45}$$

$$x^2 = 180 \cdot 45$$

$$x = \pm \sqrt{8,100} = \pm 90.$$

The prey is moving forward, so assume $x = 90$.



Then

$$\begin{aligned} L &= \int_0^{90} \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{1/2} dx \\ &= \int_0^{90} \left(1 + \frac{4x^2}{45^2} \right)^{1/2} dx \\ &\approx 209 \text{ m} \end{aligned}$$

since

$$\int (1 + kx^2)^{1/2} dx$$

$$= \frac{1}{2} x \sqrt{1 + kx^2} + \frac{1}{2\sqrt{k}} \log(\sqrt{1 + kx^2} + \sqrt{k} x) + C$$

by a Trig substitution.